FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

PART TEST – II

PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	C	19.	B	37.	A
2.	D	20.	D	38.	A
3.	C	21.	D	39.	C
4.	C	22.	D	40.	C
5.	C	23.	C	41.	A
6.	B, D	24.	A, D	42.	B, C
7.	A, C	25.	B, D	43.	A, C
8.	A, D	26.	A, B, C	44.	A, B
9.	A, B, C, D	27.	A, B, C	45.	A, B, C, D
10.	A, D	28.	B, C, D	46.	A, B
11.	A, B, C, D	29.	A, B, C, D	47.	A, B
12.	A, C, D	30.	B, C, D	48.	A, B, C
13.	B, C	31.	A, B, C, D	49.	B, D
14.	3	32.	5	50.	5
15.	0	33.	0	51.	4
16.	1	34.	8	52.	2
17.	8	35.	3	53.	1
18.	2	36.	5	54.	6

Physics

PART – I

SECTION – A

1. $\vec{F} = q\vec{v} \times \vec{B}$

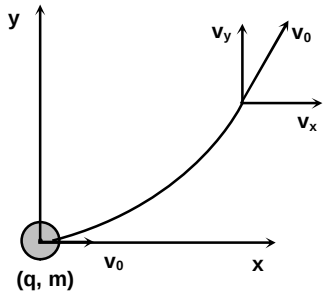
$= q(v_x \hat{i} + v_y \hat{j})\vec{B}$

$F_y = q\frac{B_0}{\ell^2}x^2 v_x$

$\int_0^{v_0} dv_y = \frac{qB_0}{m\ell^2}\int_0^{x_{max}} x^2 dx$

$v_0 = \frac{qB_0}{m\ell^2}\frac{x_{max}^3}{3}$

$x_{max} = \left(\frac{3m\ell^2 v_0}{qB_0}\right)^{1/3}$



Alternate solution

Impulse momentum theorem along x-axis

$\int_0^{x_{max}} \frac{qB_0 x^2}{\ell^2} v_x dt = mv_0$

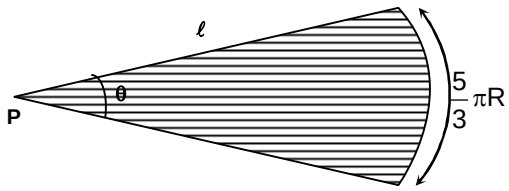
$\therefore \quad \int_0^{x_{max}} \frac{qB_0 x^2}{\ell^2} dx = mv_0$

$\therefore \; x_{max} = \left(\frac{3m\ell^2 v_0}{qB_0}\right)^{1/3}$

2. Potential due to circular non conducting disc, $V_C = \frac{\sigma r}{2\varepsilon_0}$ (where r is radius of disc)

So, $v_P = \left(\frac{\theta}{2\pi}\right)\left(\frac{\sigma \cdot \ell}{2\varepsilon_0}\right)$

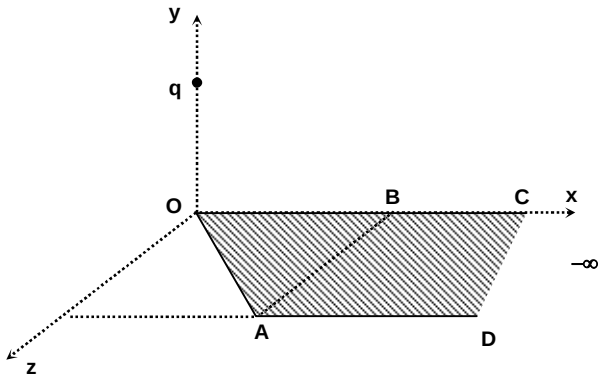
$= \left(\frac{\sigma\ell}{2\varepsilon_0}\right)\left(\frac{5}{3}\frac{\pi R}{(\ell)}\right)\left(\frac{1}{2\pi}\right) = \frac{5\sigma R}{12\varepsilon_0}$



3. Work performed = $\int_R^{2R} \frac{1}{2}\varepsilon_0 E^2 dV + \int_{3R}^{4R} \frac{1}{2}\varepsilon_0 E^2 dV = \frac{7}{96}\frac{q^2}{\pi\varepsilon_0 R}$

4. $\phi = \phi_{OAB} + \phi_{ABCD}$

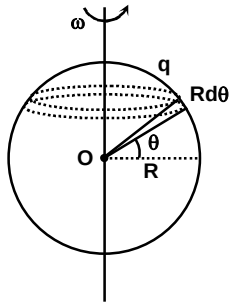
$$= \frac{q}{48\varepsilon_0} + \frac{q}{48\varepsilon_0} = \frac{q}{24\varepsilon_0}$$



5. Magnetic field due to the ring on its axis is

$$B = \frac{\mu_0 \pi r^2 i}{2\pi(r^2 + x^2)^{3/2}} \text{ (due to ring)}$$

$$B = \int_{-\pi/2}^{+\pi/2} \frac{\mu_0 \pi (R\cos\theta)^2 \left(\frac{2\pi R\cos\theta R d\theta \sigma}{2\pi/\omega}\right)}{2\pi\left[(R\cos\theta)^2 + (R\sin\theta)^2\right]^{3/2}}$$



Where $\sigma = \frac{q}{4\pi R^2}$

$$B = \frac{\mu_0 q\omega}{6\pi R}$$

6. By applying Kirchhoff's law charge transferred by the battery is $\frac{80}{19}$C.

7. Time between two consecutive collision = $\frac{1}{1000}$ sec

So, $\frac{2\ell}{v_{rms}} = \frac{1}{1000}$

$v_{rms} = 2000 \times 5 = 10000$ m/s

$$v_{rms} = \sqrt{\frac{3RT}{M}} = 10000$$

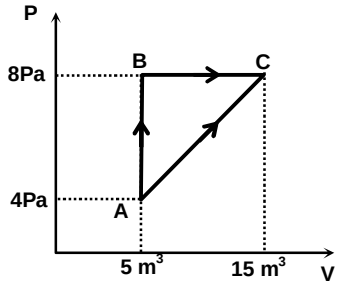
So, $T = \frac{10^8 \times 4 \times 10^{-3}}{3 \times \left(\frac{25}{3}\right)} = 16 \times 10^3 K$

Mass of helium gas $\Rightarrow$ PV = nRT

$PV = \frac{m}{M}RT$

$$m = \frac{PVM}{RT} = \frac{320 \times 125 \times (4 \times 10^{-3})}{\left(\frac{25}{3}\right) \times 16 \times 10^3} = 1.2\,gm$$

8. In path A→C



$dW = 4\times10 + \frac{1}{2}\times10\times4 = 40 + 20 = 60$ J

dQ = dU + dW

$200 = U_C - U_A + 60$

$U_C = 140 + U_A = 140 + 10 = 150$ J

In path A → B

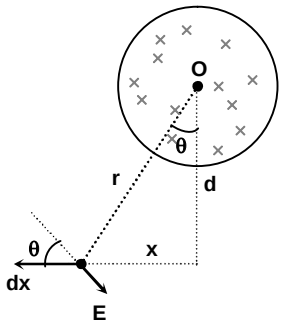
$Q_{AB} = U_B - U_A$

$= U_B\left(1-\frac{U_A}{U_B}\right) = U_B\left(1-\frac{T_A}{T_B}\right)$

$= 20\left(1-\frac{1}{2}\right) = 10$ Joule

9. $\int\vec{E}\cdot d\vec{\ell} = A\cdot\frac{dB}{dt}$

$E\cdot2\pi\sqrt{x^2+d^2} = \pi R^2k$, where $(\frac{dB}{dt}=k)$



$E = \frac{R^2k}{2\sqrt{x^2+d^2}}$

$W = \int_0^\infty q\vec{E}\cdot d\vec{x}$

$= \int_0^\infty qE\cos\theta\cdot dx = \int_0^\infty \frac{qR^2k}{2\sqrt{x^2+d^2}}\frac{d}{\sqrt{d^2+x^2}}dx = \frac{qR^2k\pi}{4}$

Alternate solution

By applying Faraday's Law for an imaginary triangular loop having vertex O, P and point at infinity, $q\int\vec{E}\cdot d\vec{\ell}$ can be found out.

10. $\phi = BA = (t^2)(\pi r^2)$

$\varepsilon = \frac{d\phi}{dt} = 2\pi r^2t$

$i = \frac{\varepsilon}{R} = \frac{2\pi r^2t}{R}$

For toppling

$\tau = |\vec{\mu}\times\vec{B}| \geq mgr$

$(i.\pi r^2).1 \geq mgr$

$\frac{2\pi r^2t}{R}\cdot\pi r^2 \geq mgr$

$t \geq \frac{mgR}{2\pi^2r^3} = 1$sec

Heat $= \int i^2Rdt = \int\left(\frac{2\pi r^2t}{R}\right)^2 dt.R$

Heat $= \frac{4\pi^2r^4}{R}\int_0^1 t^2dt = \frac{80}{3}$Joule

11. $V_L = L\dfrac{di}{dt} \Rightarrow Ldi = V_L\,dt$

$L\int di = \int V_L dt = \dfrac{1}{2}\cdot 6\cdot 10 = 30$

$L(i_f - i_i) = 30$ Amp

$i_f = \dfrac{30}{10} = 3$ amp

for $t \le 2$,

$V_L = \dfrac{10}{2}t = 5t$

$\int_0^i Ldi = \int V_L dt = 5\int_0^1 tdt$

$L(i-0) = \dfrac{5}{2}$

$\Rightarrow i = 1/4$

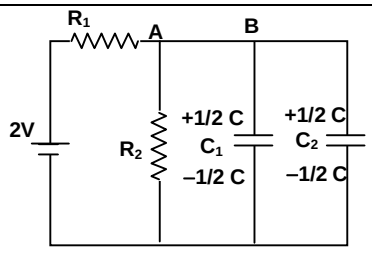
for $t \ge 2$

$V_L = \left(15 - \dfrac{5}{2}t\right)$

$\int_1^i Ldi = \int_2^4 \left(15 - \dfrac{5}{2}t\right)dt$

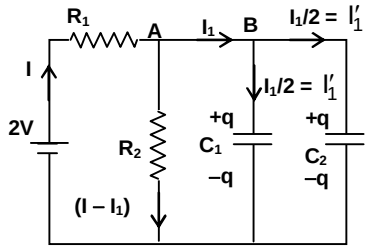
$\Rightarrow i = 5/2$ amp

12.



(at t = 0)

When the switch is closed then at t = 0 the distribution of the charge is shown in the figure.



By applying Kirchhoff's Law $I_1' = \dfrac{(1-q)}{2}$

$\int_{1/2}^{q} \dfrac{dq}{1-q} = \int_0^t \dfrac{dt}{2}$

$q = \left(1 - \dfrac{1}{2}e^{-t/2}\right)$

$I_1' = \dfrac{dq}{dt} = \dfrac{1}{4}e^{-t/2}$

13. $H_1 = \dfrac{V^2}{R_1}(30\times 60)$, $H_2 = \dfrac{V^2}{R_2}(45\times 60)$

$H_1 = H_2 \Rightarrow \dfrac{30}{R_1} = \dfrac{45}{R_2} \Rightarrow 3R_1 = 2R_2$

$$H_{series} = \frac{V^2}{(R_1+R_2)}t_s = \frac{V^2}{\left(R_1+\frac{3}{2}R_1\right)}t_s = H_1 = \frac{V^2}{R_1}(30\times 60)$$

$\Rightarrow t_s = (30\times 60)\frac{5}{2} = 75$ min

$$H_P = \left(\frac{V^2}{R_1}+\frac{V^2}{R_2}\right)t_P = \left(\frac{V^2}{R_1}+\frac{2V^2}{3R_1}\right)t_P$$

$$\Rightarrow \frac{5V^2}{3R_1}t_P = \frac{V^2}{R_1}(30\times 60)$$

$$t_P = \frac{3\times 30\times 60}{5} = 18 \text{ min}$$

SECTION – C

14. $dQ = dU + dW$
$dQ = -2dU$
$-3dU = dW$
$-3nC_V dT = PdV$
$$\frac{-3nRdT}{\gamma-1} = \left(\frac{nRT}{V}\right)dV$$
From this $TV^{\frac{\gamma-1}{3}} = \text{constant}$
So, K = 3

15. There is no transfer of charge through the battery

16. $P_{av} = VI\cos\phi = V\left(\frac{V}{z}\right)\left(\frac{R}{z}\right) = \frac{V^2R}{Z^2}$
It means Z has to be same in both cases
$Z_1 = Z_2 \Rightarrow \sqrt{R^2+X_L^2} = \sqrt{R^2+(X_C-X_L)^2}$
$2X_L = X_C$
$$\Rightarrow C = \frac{1}{2\omega^2 L} = 10^{-6} = 1\times 10^{-6}$$

17. $v_y = u_y + a_y t$
$$0 = 6 - \frac{qE_0}{m}t_1$$
$t_1 = 3$ sec
$v_x^2 + v_y^2 + v_z^2 = (5\sqrt{5})^2$
$4^2 + v_y^2 + 3^2 = 125$
$v_y = 10$ m/sec
$v_y = u_y + a_y t$
$$10 = 0 + \frac{qE_0}{m}t_2$$
$\Rightarrow t_2 = 5$ sec
So, $t = t_1 + t_2 = 8$ sec

18. i_1 is current in the first loop

$$i_1 = \frac{40}{20}\left(1 - e^{-\frac{t}{5\times10^{-4}}}\right) = \frac{3}{2}\text{ amp}$$

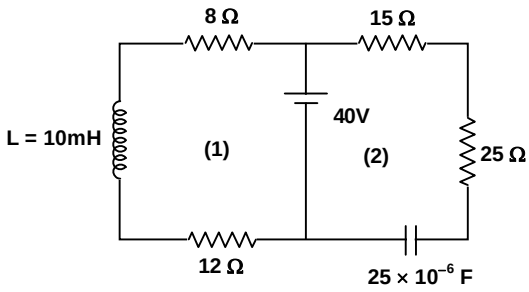
i_2 is current in the second loop

$$i_2 = \frac{40}{40}e^{-\frac{t}{10^{-3}}} = \frac{1}{2}\text{ amp}$$

So, total current through key (K)

$i = i_1 + i_2$

$$= \frac{3}{2} + \frac{1}{2} = 2\,\text{amp}$$

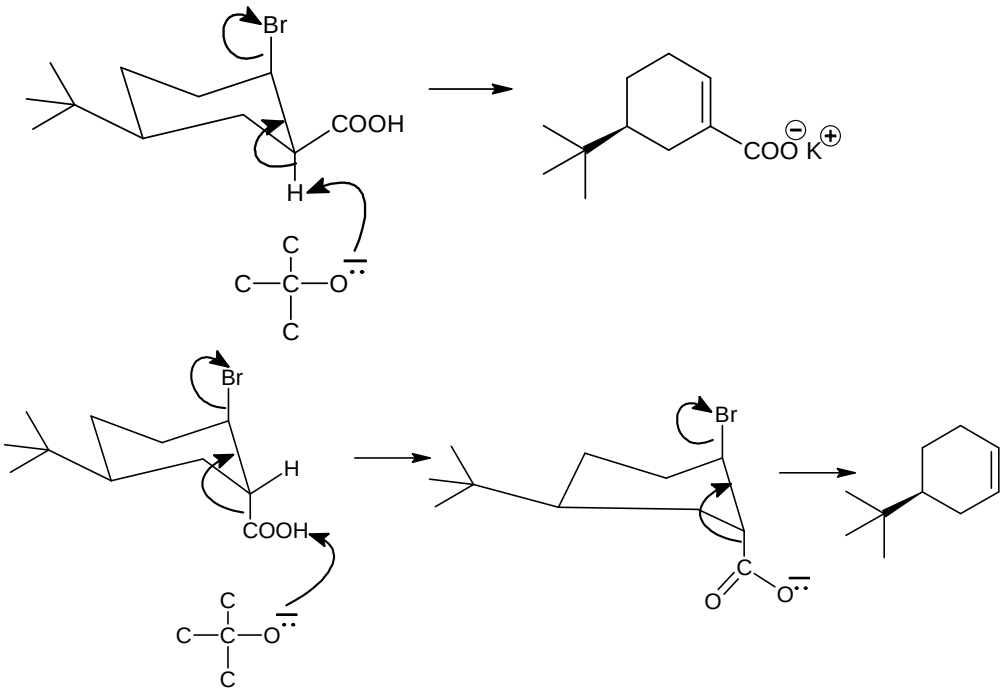


Chemistry

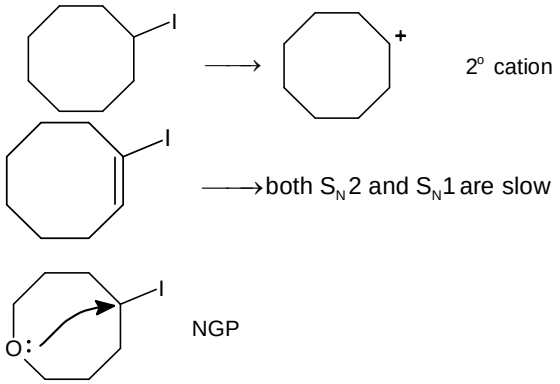
PART – II

SECTION – A

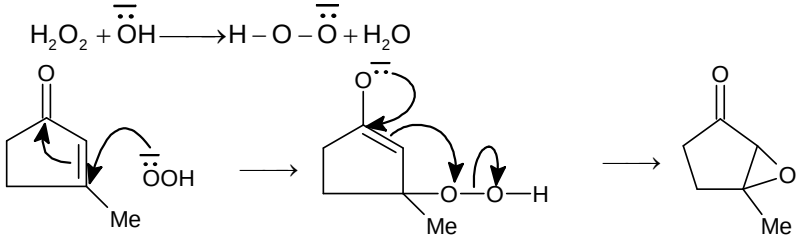
19.



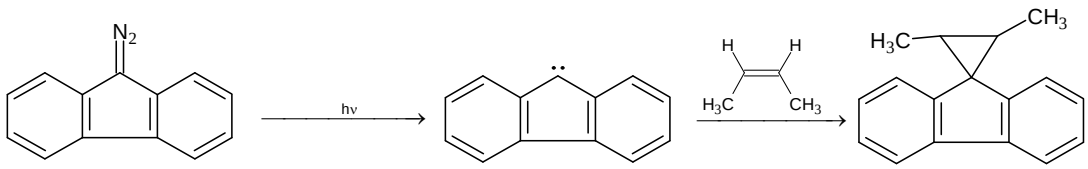
20.



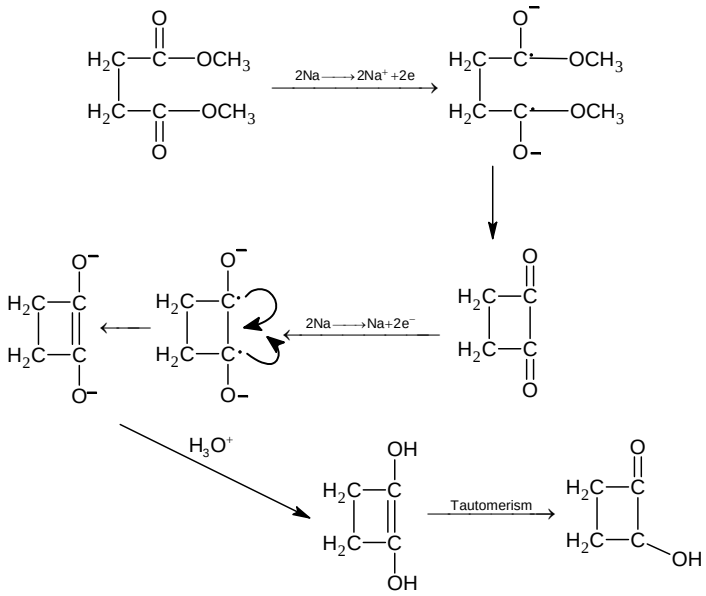
21.



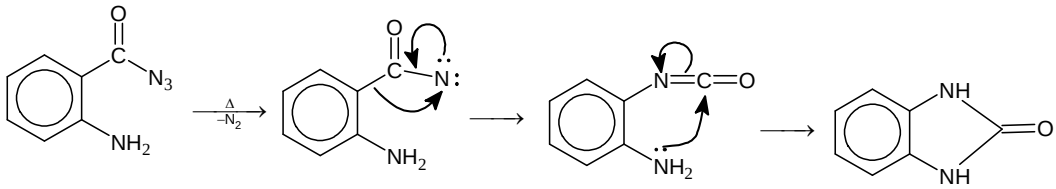
22.



23. Acyloin condensation



25.



26. $H_3N: + B_2H_6 \longrightarrow \left[H_3N \longrightarrow BH_2 \longleftarrow NH_3\right]^+ \left[BH_4\right]^-$

$CH_3NH_2 : + B_2H_6 \longrightarrow \left[CH_3NH_2 \longrightarrow BH_2 \longleftarrow NH_2CH_3\right]^+ \left[BH_4\right]^-$

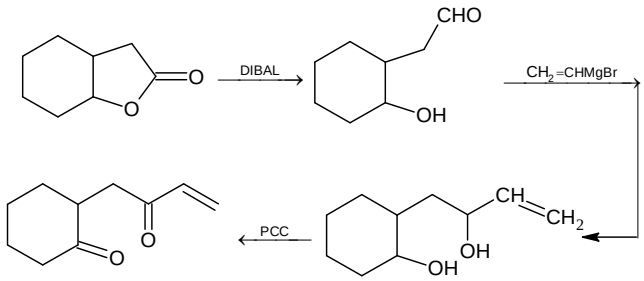
$(CH_3)NH + B_2H_6 \longrightarrow \left[(CH_3)_2NH \longrightarrow BH_2 \longleftarrow NH(CH_3)_2\right]^+ \left[BH_4\right]^-$

27. Facts

28. Facts

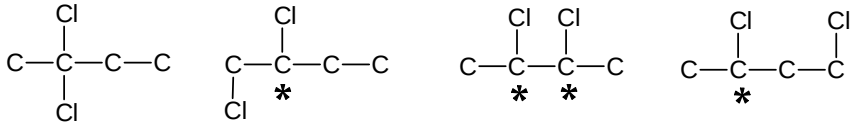
29. Glucose, Fructose, Maltose are reducing sugars.

30.

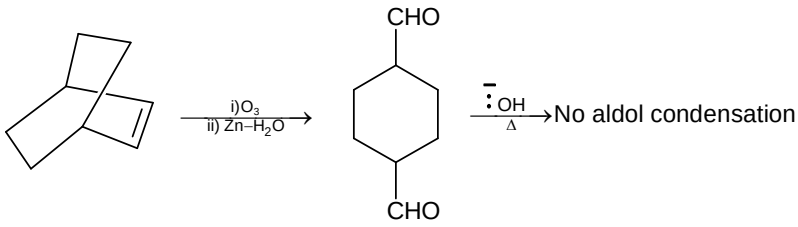


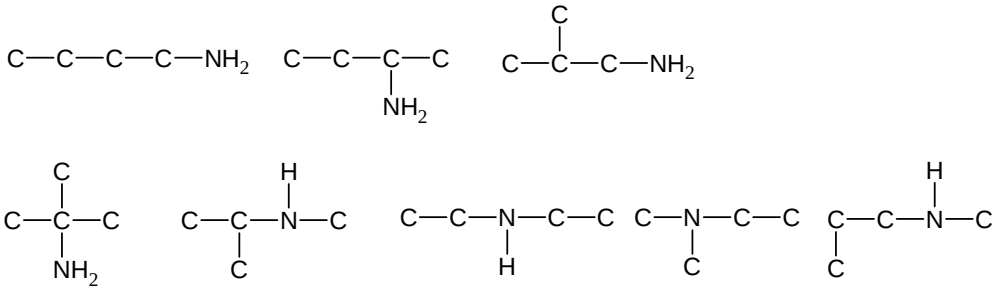
31. A, B, D because hydrate is unstable, 'C' due to steric factor.

SECTION – C

32. 

(1) (1) (2) (1)

33. 

34. 

Mathematics

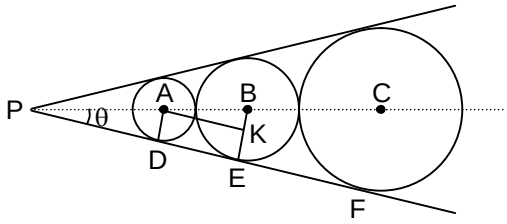
PART – III

SECTION – A

37. Let F be the fixed focus and M be the moving focus and T be the varying points of mutual tangency. The tangent line at T makes equal angle with FT and with a vertical line. This and congruence of the two parabola imply that MT is vertical and FT = MT and must line on directrix of $y = x^2 - x + 1$

So, $y = \frac{3}{4}$

38. $DE = AK = \sqrt{(r_2 + r_1)^2 - (r_2 - r_1)^2} = 2\sqrt{r_1 r_2}$



Similarly $EF = 2\sqrt{r_2 r_3}$

$\angle APD = \theta$ and $PD = x$ then

$$\tan\theta = \frac{AD}{PD} = \frac{BE}{PE} = \frac{CF}{PF} = \frac{r_1}{x}$$

$$= \frac{r_2}{x + 2\sqrt{r_1 r_2}} = \frac{r_3}{x + 2\sqrt{2r_1 r_2} + \sqrt{r_2 r_3}}$$

Hence, $\frac{r_2 - r_1}{2\sqrt{r_1 r_2}} = \frac{r_3 - r_2}{2\sqrt{r_2 r_3}}$

$\Rightarrow r_2 = \sqrt{r_1 r_3} \Rightarrow r_2 = \sqrt{2 \times 8} = 4$

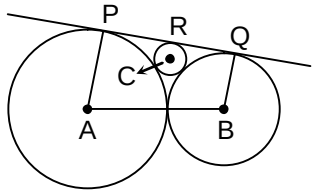
39. Since In x AM and ND are concurrent $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$a(bc - a^2) - b(b^2 - ac) + c(ab - c^2) = 0$
$abc - a^3 - b^3 + abc + abc - c^3$
$3abc - a^3 - b^3 - c^3 = 0$
$a^3 + b^3 + c^3 = 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega) = 0$

40. Solve y = x with circle
Circle must passes through the foci of the ellipse

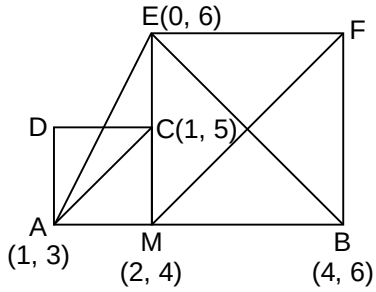
41. Using length of direct common tangent

$$\left.\begin{aligned} PQ^2 &= 4r_1 r_2 \\ QR^2 &= 4r_1 r_3 \end{aligned}\right| = PQ = PR + QR$$



$$\Rightarrow \frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$$

42. Shown in the figure since C(1, 5) is the orthocentre of triangle ΔAEB, similarly for the other side the coordinates of C(3, 3)



43. Given circle is $(x-1)^2 + (y-3)^2 = 1$

Let of tangent to circle is $y - 3 = m(x-1) + 1\sqrt{1+m^2}$

(3, 4) lies on axis

$1 = 2m + \sqrt{1+m^2}$

$(1-2m)^2 = 1 + m^2$

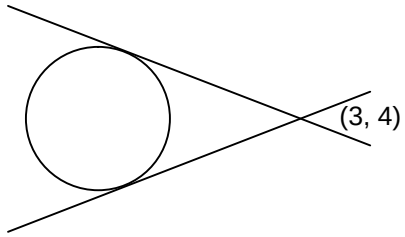
$4m^2 - 2m + 1 = 1 + m^2$

$3m^2 - 4m = 0$

$m = 0, \frac{4}{3}$

$\Rightarrow \frac{y-4}{n-3} = m = 0.0r\frac{4}{3}$

Smallest is 0 and tangent = $\frac{4}{3}$

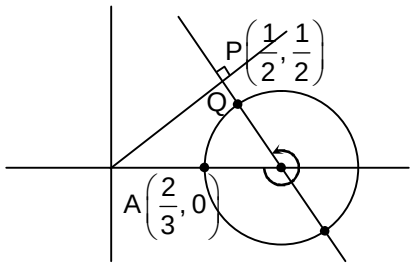


44. Eg. of common normal $y - 0 = -1(x-1)$

$y = -x + 1$ $\qquad (x-1)^2 + y^2 = \frac{1}{3}$

$x + y - 1 = 0$ $\qquad 2y^2 = \frac{1}{9}$

$x - 1 = y$ $\qquad y^2 = \frac{1}{3\sqrt{2}}$



Clearly shortest distance of approach is along common normal whose equation is $x + y - 1 = 0$

So when they are at their closest distance P must be at $\left(\frac{1}{2}, \frac{1}{2}\right)$ and Q must be at

$\left(1-\frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}\right)$ covering angle $\frac{7\pi}{4}$ on the circle

Since P and Q happened to be at these points at same time so we can have

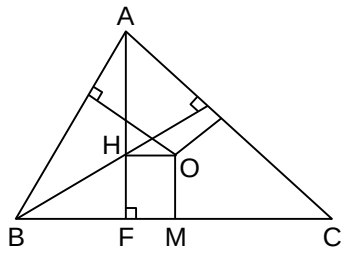
$t_0 = \frac{7\pi}{4w} = \frac{1}{\sqrt{2}u}$ also at this instance

t_0 , $r_{min} = \frac{3-\sqrt{2}}{3\sqrt{2}} \Rightarrow [12r_{min}] = \left[6\sqrt{2} - 4\right] = 4$

Also at any time instant t_1 co-ordinate of P, Q are $\left(\frac{ut}{\sqrt{2}}, \frac{ut}{\sqrt{2}}\right)$ at $Q\left(1+\frac{\cos(\omega t)}{3}, +\sin\left(\frac{\omega t}{3}\right)\right)$

$\therefore\ r^2(t) = \left(\frac{ut}{\sqrt{2}} - 1 - \frac{\cos(\omega t)}{3}\right)^2 + \left(\frac{ut}{\sqrt{2}} - \frac{\sin(\omega t)}{3}\right)^2$

45. Centroid G of triangle is collinear with H and O and G lies two third of way from A to M, therefore H is two third of the way from A to F. So AF = 3 × OM = 15



Since $\angle HFB = \angle AFC$, $\angle HBF = \angle ACF$

So, ΔBFH and ΔAFC are similar

$\frac{BF}{HF} = \frac{AF}{FC} \Rightarrow BF \cdot FC = FA \cdot HF = 75$

Now $BC^2 = (BF + CF)^2 = (BF - CF)^2 + 4BF \cdot FC$

But FC – BF = (FM + MC) – (BM – FM) = 2FM = 22

$BC = \sqrt{22^2 + 4 \times 75} = 28$

46. $\frac{\frac{\sqrt{3}-1}{2\sqrt{2}}}{\sin x} + \frac{\frac{\sqrt{3}+1}{2\sqrt{2}}}{\cos x} = 2$

$\Rightarrow \sin\frac{\pi}{12} \cdot \cos x + \cos\frac{\pi}{12} \sin x = \sin 2x$

$\Rightarrow \sin\left(x + \frac{\pi}{12}\right) = \sin 2x$

$\Rightarrow n = \frac{\pi}{12}$ and $\frac{11\pi}{36}$

47. Since $\frac{\frac{a}{b}-1}{\frac{a}{b}+1} = \tan\frac{A-B}{2}\tan\frac{C}{2}$

$\Rightarrow A - B = 90^\circ$ and $A + B = 120^\circ$

$\Rightarrow A = 105^\circ$ and $B = 15^\circ$

48. $x^2 = \cos^2 A + \cos^2 B + 2 \cos A \cdot \cos B$

$y = 2(\cos^2 A + \cos^2 B) - 2$

$\therefore \cos^2 A + \cos^2 B = 1 + \frac{y}{2}$

$\because \cos A \cdot \cos B = \frac{1}{4}\left(2x^2 - y - 2\right)$ and $z = -2x^3 + 3xy + 3x$

$\therefore 2x^3 + z = 3x(y + 1)$

$xyz = 0 \ \forall$ A and B is not true

49. P lies on circumcircle

So, L, M, N are collinear (i.e. Simson's line)

SECTION – C

50. Use $\sin^3 x = \frac{3\sin x - \sin 3x}{4}$ and $\cos^3 x = \frac{3\cos x + \cos 3x}{4}$

Equation becomes $\sin 4x = 0$

$x = \frac{n\pi}{4}$

51. Put $\sin 2\theta = x$ given equation becomes $x^4 + 1 = 17(1 + 2x + x^2)^2$

Divide by x^2, we have $x^2 + \frac{1}{x^2} = 17\left(x + \frac{1}{x} + 2\right)$

Now put $y = x + \frac{1}{x} \Rightarrow y = -\frac{7}{4}$ or $-\frac{5}{2}$

Now $y = -\frac{7}{4}$ have no solution and $y = -\frac{5}{2}$, we have $x = -\frac{1}{2}, -2$

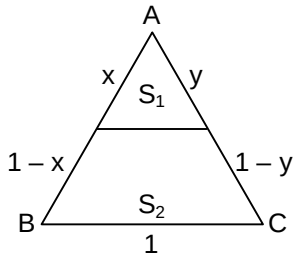
So, there are (4) solution in $[0, 2\pi]$

52. $x + y = 1 - x + 1 - y$ 1

$x + y = \frac{3}{2}$

$S_3 = S_1 + S_2$

$\frac{S_1}{S_3} = \frac{\frac{1}{2}xy\sin 60^o}{\frac{\sqrt{3}}{4}} = \frac{xy\sqrt{3}}{\left(\frac{\sqrt{3}}{4}\right)} = xy = x\left(\frac{3}{2} - x\right)$



$\because S_2 = S_3 - S_1$

$\frac{S_2}{S_3} = 1 - \frac{S_1}{S_3} = 1 - xy = 1 - x\left(\frac{3}{2} - x\right)$

$\frac{S_1}{S_3}(\max) = \frac{3}{16}$ as $\frac{S_2}{S_3} = \frac{16}{9} - 1 = \left(\frac{7}{9}\right)$

53. Equation of tangent $\frac{xh}{a^2} - \frac{2y}{b^2} - 1 = 0$; $x = \left(1 + \frac{2y}{b^2}\right)\frac{a^2}{h}$

By solving $\left(\left(1 + \frac{2y}{b^2}\right)\frac{a^2}{h}\right)^2 + y^2 - a^2 = 0$

$\left(1 + \frac{4y}{b^2} + \frac{4y^2}{b^4}\right)\frac{a^4}{h^2} + y^2 - a^2 = 0$

$y^2\left(\frac{4a^4}{b^4h^2} + 1\right) + \frac{4}{b^2} \cdot \frac{a^4}{h^2} \cdot y + \frac{a^4}{h^2} - a^2 = 0$

$\Rightarrow \frac{y_1 + y_2}{y_1 y_2} = 1$

54. $\because rr_1r_2r_3 = \Delta^2$